



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**SENIOR CERTIFICATE EXAMINATIONS/
NATIONAL SENIOR CERTIFICATE EXAMINATIONS
SENIORSERTIFIKAAT-EKSAMEN/
NASIONALE SENIORSERTIFIKAAT-EKSAMEN**

MATHEMATICAL LITERACY P2/WISKUNDIGE GELETTERDHEID V2

2022

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

Symbol/Kode	Explanation/Verduideliking
M	Method/Metode
MA	Method with accuracy/Metode met akkuraatheid
CA	Consistent accuracy/Volgehoue akkuraatheid
A	Accuracy/Akkuraatheid
C	Conversion/Herleiding
S	Simplification/Vereenvoudiging
RT	Reading from a table/graph/document/diagram/Lees vanaf tabel/grafiek/dokument/diagram
SF	Correct substitution in a formula/Korrekte vervanging in 'n formule
O	Opinion/Explanation/Opinie/Verduideliking
P	Penalty, e.g. for no units, incorrect rounding off, etc./Penalisasie, bv. vir geen eenhede, verkeerde afronding, ens.
R	Rounding off/Afronding
NPR	No penalty for correct rounding/Geen penalisasie vir korrekte afronding nie
AO	Answer only/Slegs antwoord
MCA	Method with consistent accuracy/Metode met volgehoue akkuraatheid
RCA	Rounding consistent with accuracy/Afronding met volgehoue akkuraatheid
*	Asterisk means refer to attached notes

**These marking guidelines consist of 19 pages.
Hierdie nasien riglyne bestaan uit 19 bladsye.**

NOTE:

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- If a candidate has crossed out (cancelled) an attempt to a question and NOT redone the solution, mark the crossed out (cancelled) version.
- Consistent accuracy (CA) applies in ALL aspects of the marking guidelines; however it stops at the second calculation error.
- Note: Consistent accuracy (CA) does NOT apply in cases of a breakdown.
- If the candidate presents any extra solution when reading from a graph, table, layout plan and map, then penalise for every extra item presented.
- As a general marking principle, if a candidate has incurred one mistake and there is evidence of sound mathematics thereafter, then that candidate should lose ONE mark only.

LET WEL:

- As 'n kandidaat 'n vraag TWEE KEER beantwoord, sienslegs die EERSTE poging na.
- As 'n kandidaat 'n antwoord van 'n vraag doodtrek (kanselleer) en nie oordoen nie, sien die doodgetrekte (gekanselleerde) poging na.
- Volgehoue akkuraatheid (CA) word in ALLE aspekte van die nasienriglyne toegepas, dit hou op by die tweede berekeningsfout.
- Let wel: Volgehoue akkuraatheid (CA) geld NIE in die geval van 'n afbreuk NIE.
- Wanneer 'n kandidaat aflesings vanaf 'n grafiek, tabel, uitlegplan en kaart geneem het en ekstra antwoorde gee, penaliseer vir elke ekstra item.
- 'n Algemene nasienbeginsel is dat, indien 'n kandidaat een fout maak en daarna voortgaan met korrekte wiskunde, die kandidaat slegs EEN punt verloor.

QUESTION/VRAAG 1 [30 MARKS/PUNTE] – ANSWER ONLY ACCEPTED			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
1.1.1	D ✓✓ A	2A correct option Accept 1:50 000 (2)	MP L1
1.1.2	E ✓✓ A	2A correct option (2)	M L1
1.1.3	G ✓✓ A	2A correct option Accept 1 cm = 1 m (2)	MP L1
*1.1.4	C ✓✓ A	2A correct option (2)	M L1
*1.1.5	F ✓✓ A	2A correct option (2)	M L1
1.2.1	✓✓ A B OR/OF (2 × 240 × 70 + 2 × 240 × 112 + 2 × 112 × 70) mm ²	2A correct option (2)	M L1
*1.2.2	✓✓ A mm ³ OR Cubic millimetres/Kubieke millimeter	2A correct unit (2)	M L1
1.2.3	✓ C Length/Lengte = 240 ÷ 1 000 = 0,24 m ✓ A	1C conversion factor 1A simplification (2)	M L1

Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
*1.2.4	Number of rows/ <i>Getal rye</i> $= \frac{2\,100\text{ mm}}{70\text{ mm}} \checkmark \text{ A}$ $= 30 \checkmark \text{ CA}$	1A height 1 A correct denominator 1CA number of rows (3)	MP L1
1.3.1	Mass of the flour (in kg)/ <i>Massa van die meel</i> $= \frac{500}{1\,000} \checkmark \text{ C}$ $= \frac{1}{2} \text{ kg or/of } 0,5 \text{ kg} \checkmark \text{ A}$	1C divide by 1 000 1A simplification (2)	M L1
1.3.2	$\checkmark \text{ A}$ 12 scones/ <i>botterbroodjies</i> = 2 eggs/ <i>eiers</i> 6 scones/ <i>botterbroodjies</i> = 1 egg/ <i>eier</i> 30 scones = 2 + 2 + 1 = 5 eggs/ <i>eiers</i> $\checkmark \text{ A}$ OR/OF $\checkmark \text{ A}$ 12 scones/ <i>botterbroodjies</i> = 2 eggs/ <i>eiers</i> 30 scones/ <i>botterbroodjies</i> = $\frac{30}{12} \times 2$ $= 5 \text{ eggs/eiers} \checkmark \text{ A}$ OR/OF 30 scones/ <i>botterbroodjies</i> = $\frac{30}{12} = 2,5 \text{ dozen/dosyn} \checkmark \text{ A}$ 1 dozen need 2 eggs/1 <i>dosyn benodig 2 eiers</i> 2,5 dozen/ <i>dosyn</i> = $2,5 \times 2 = 5 \text{ eggs/eiers} \checkmark \text{ A}$	1A dozen = 12 1A simplification OR/OF 1A dozen = 12 1A simplification OR/OF 1A dozen = 12 1A simplification (2)	M L1
1.3.3	Radius = $7 \text{ cm} \div 2 \checkmark \text{ MA}$ $= 3,5 \text{ cm OR/OF } 35 \text{ mm} \checkmark \text{ A}$	1MA dividing by 2 1A radius (2)	M L1
1.3.4	Number of dozen scones/ <i>Getal dosyn botterbroodjies</i> $= \frac{500}{75} \checkmark \text{ MA}$ $= 6,67 \checkmark \text{ S}$ $= 6 \checkmark \text{ R}$	1MA dividing by 75 1S simplification 1R rounding down (3)	M L1
*1.3.5	$\checkmark \text{ A}$ $\checkmark \text{ A}$ Ten minutes past two in the afternoon. <i>Tien minute oor twee in die namiddag.</i>	1A time 1A afternoon (2)	M L1
[30]			

QUESTION/VRAAG 2 [32 MARKS/PUNTE]			
Q/V	Solution/Oplissing	Explanation/Verduideliking	T/L
2.1.1	14:00 ✓✓ A	2A 24-hour time format (2)	MP L1
2.1.2	8 ✓✓ A	2A correct number (2)	MP L2
2.1.3	Bicycle/Fiets ✓✓ RT	2RT bicycle (2)	MP L1
2.1.4(a)	Distance/Afstand = 9K + 1K = 10 km ✓✓ A Fraction/Breuk = $\frac{10}{42,2}$ ✓ MCA = $\frac{50}{211}$ ✓ CA OR/OF Distance/Afstand = 1 000 × 10 = 10 000 m ✓✓ A 42,2 km = 42 200 m Fraction/Breuk = $\frac{10\,000\,m}{42\,200\,m}$ ✓ MCA = $\frac{50}{211}$ ✓ CA	2A 10 km 1MCA correct order 1CA simplification OR/OF 2A 10 km 1MCA correct order 1CA simplification (4)	MP L2
2.1.4 (b)	✓✓ A The distance is less than a full marathon. Die afstand is minder as 'n vol marathon. OR/OF It is shorter than a standard marathon. Dit is korter as 'n standaard marathon. OR/OF It is a fraction of a full marathon. Dit is 'n breuk van die vol marathon.	2A explanation (2)	MP L4
2.1.5	B ✓✓ A OR/OF 0%	2A correct option (2)	P L2
2.2.1	Scout House/Verkennerhuis ✓✓ RT	2RT correct place of interest (2)	MP L2
*2.2.2	South-east OR SE ✓✓ A Suidoos OF SO	2A correct direction (2)	MP L1

Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
*2.2.3	Nellmapius ✓✓RT	2RT correct street (2)	MP L2
2.2.4	St Martin's Church/ <i>St Martin-Kerk</i> ✓✓RT	2RT correct church (2)	MP L2
2.2.5	Irene Library & Hall/ <i>Irene Biblioteek & Saal</i> ✓✓✓RT [Accept Hall / <i>Aanvaar Saal</i>]	3RT correct place (3)	MP L3
2.2.6	Measured distance/ <i>Gemete afstand</i> = 8 cm ✓MA 8 cm : 1,9 km ✓MCA 8 cm : 190 000 cm ✓C Scale/ <i>Skaal</i> is 1 : 23 750 ✓S 1 : 24 000 ✓R (Maximum distance/ <i>maksimum afstand</i>) Measured distance/ <i>Gemete afstand</i> = 8,4cm ✓MA 8,4 cm : 1,9 km ✓MCA 8,4 cm : 190 000 cm ✓C Scale/ <i>Skaal</i> is 1 : 22 619,05 ✓S 1 : 23 000 ✓R OR/OF ✓MA ✓C 8,4cm ÷ 100 000 : 1,9 km ✓MCA 0,000084 km : 1,9km 1: 22 619 ✓S 1: 23 000 ✓R	1MA correct measurement 1MCA correct ratio 1C converting km to cm 1S simplified ratio 1R correct rounding OR/OF 1MA correct measurement 1C converting cm to km 1MCA correct ratio 1S simplified ratio 1R correct rounding Provinces need to mark according to ± 1 mm of their provincial paper. (5)	MP L3

Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
2.2.7	The traffic flow is in the opposite direction. ✓✓O <i>Die verkeervloei in die teenoorgestelderigting.</i> OR/OF One-way traffic /The arrow shows you can only turn left. <i>Eenrigtingverkeer/ Die pyl wys jy kan slegs links draai</i> OR/OF The driver will be facing oncoming traffic. <i>Die bestuurder sal in aankomende verkeer inry.</i>	2O opinion (2)	MP L4
		[32]	

QUESTION/VRAAG 3[29MARKS/PUNTE]			
Q/V	Solution/Oplissing	Explanation/Verduideliking	T/L
3.1.1	Total length/<i>Totale lengte</i> $= 55 \text{ cm} + 99 \text{ cm} + 55 \text{ cm} = 209 \text{ cm}$ ✓A Perimeter/<i>Omtrek</i> $= 2(209 \text{ cm} + 149 \text{ cm})$ ✓SF $= 2(358) \text{ cm}$ $= 716 \text{ cm}$ ✓CA OR/OF Perimeter/<i>Omtrek</i> $= (149 + 55 + 99 + 55 + 149 + 55 + 99 + 55) \text{ cm}$ ✓A ✓SF $= 716 \text{ cm}$ ✓CA OR/OF Perimeter/<i>Omtrek</i> $= 2(149) \text{ cm} + 2(55 + 99 + 55) \text{ cm}$ ✓A ✓SF $= (298 + 418) \text{ cm}$ $= 716 \text{ cm}$ ✓CA	1A total length 1SF substitution 1CA perimeter OR/OF 1A total length 1SF substitution 1CA perimeter OR/OF 1A total length 1SF substitution 1CA perimeter (3)	M L2
3.1.2	Radius $= \frac{605}{2} = 302,5 \text{ mm}$ ✓A $= 30,25 \text{ cm}$ ✓C Area/<i>Oppervlakte</i> $= 3,142 \times (30,25 \text{ cm})^2$ ✓SF $= 2\,875,126375 \text{ cm}^2$ ✓CA OR/OF Radius $= \frac{605}{2} = 302,5 \text{ mm}$ ✓A Area/<i>Oppervlakte</i> $= 3,142 \times (302,5 \text{ mm})^2$ ✓SF $= 28\,512,6375 \text{ mm}^2$ $= 28\,512,6375 \div 10^2$ ✓C $= 2\,875,126375 \text{ cm}^2$ ✓CA	1A radius 1C conversion 1SF substitution 1CA simplification OR/OF 1A radius 1SF substitution 1C conversion 1CA simplification NPR (4)	M L2

Q/V	Solution/Oplissing	Explanation/Verduideliking	T/L
3.1.3	$P = \frac{3}{7} \quad \checkmark A$ $= 0,4285714286 \quad \checkmark CA$ OR/OF $P = 1 - \frac{4}{7} = \frac{3}{7} \quad \checkmark A$ $= 0,4285714286 \quad \checkmark CA$	1A numerator 1A denominator 1CA decimal form OR/OF 1M subtracting from 1 1A simplification 1CA decimal form NPR (3)	P L2
*3.2.1	Total area/Totale oppervlakte $= 4 \text{ m} \times 5 \text{ m} + 3 \text{ m} \times 4 \text{ m} \quad \checkmark M$ $= 20 \text{ m}^2 + 12 \text{ m}^2 \quad \checkmark SF$ $= 32 \text{ m}^2$	1SF substitution of correct values 1M adding NPU (2)	M L2
*3.2.2 TR	Area of 1 tile/Opp van 1 teël = $35 \text{ cm} \times 35 \text{ cm} \quad \checkmark SF$ $= 1\,225 \text{ cm}^2$ $= 1\,225 \div (100)^2 \text{ m}^2 \quad \checkmark C$ $= 0,1225 \text{ m}^2 \quad \checkmark CA$ Number of tiles needed/Getal teëls nodig $= \frac{32}{0,1225} \quad \checkmark MCA$ $= 261,2244898 \quad \checkmark CA$ Number to add/Getal om by te tel $= 10\% \times 261,2244898 = 26,12244898 \quad \checkmark MCA$ Total number of tiles/Totale aantal teëls $= 261,2244898 + 26,12244898 = 287,3469388 \quad \checkmark CA$ Number of boxes/Getal bokse $= \frac{287,3469388}{4} = 71,83673469 \quad \checkmark MCA$ $\therefore 72 \text{ boxes} \quad \checkmark CA$	1 SF substitution 1C conversion 1CA simplification 1MCA dividing areas 1CA simplification 1MCA calculation 10% 1CA simplification 1MCA dividing by 4 1CA rounding up 3 marks area of tile 2 marks number of tiles 2 marks adding 10% tiles or area 2 marks number of boxes	M L3

3.2.2	OR (when rounding consistently up) /OF Area of 1 tile/<i>Opp van 1 teël</i> = $35 \text{ cm} \times 35 \text{ cm}$ ✓SF $= 1\,225 \text{ cm}^2$ $= 1\,225 \div (100)^2 \text{ m}^2$ ✓C $= 0,1225 \text{ m}^2$ ✓CA Number of tiles needed/<i>Getal teëls nodig</i> $= \frac{32}{0,1225}$ ✓MCA $= 261,2244898 \approx 262$ ✓CA Number to add/<i>Getal om by tetel</i> $= 10\% \text{ of } 262 = 26,2$ ✓MCA Total number of tiles/<i>Totale aantal teëls</i> $= 262 + 26,2 = 288,2 \approx 289$ ✓CA Number of boxes/<i>Getal bokse</i> = $\frac{289}{4}$ ✓MCA $\therefore 73 \text{ boxes}$ ✓CA OR/OF ✓C ✓SF ✓CA Area of 1 tile/<i>Opp van 1 teël</i> = $(0,35)^2 = 0,1225 \text{ m}^2$ Area covered by tiles in a box/ <i>Opp. wat 'n boks teëls bedek</i> $= 0,1225 \text{ m}^2 \times 4 = 0,49 \text{ m}^2$ ✓MCA ✓CA Area to be tiled/<i>Opp wat geteël word</i> $32 \times 110\% = 35,2$ ✓MCA ✓CA Number of boxes needed/<i>Getal bokse nodig</i> $= \frac{35,2}{0,49}$ ✓MCA $\approx 71,8$ $\therefore 72 \text{ boxes}$ ✓CA	OR/OF 1 SF substitution 1C conversion 1CA simplification 1MCA dividing areas 1CA simplification 1MCA calculation 10% 1CA simplification 1MCA dividing by 4 1CA rounding up OR/OF 1C conversion 1 SF substitution 1CA simplification 1MCA multiplying by 4 1CA simplification 1MCA calculation 10% 1CA simplification 1MCA dividing areas 1CA rounding up	
-------	---	--	--

Q/V	Solution/Oplissing	Explanation/Verduideliking	T/L
	OR/OF Area of tile/ <i>Opp van 'n teël</i> = $(35\text{cm})^2$ $= 1\,225\text{ cm}^2$ ✓A ✓A $32\text{ m}^2 \times 100^2 = 320\,000\text{ cm}^2$ ✓C Number of tiles needed/ <i>Getal teëls nodig</i> $32\text{ m}^2 = 320\,000\text{ cm}^2 \div 1\,225\text{ cm}^2$ ✓MCA $= 261,2244898$ ✓CA With extras/ <i>Met ekstras</i> = $261,2244898 \times 1,1$ MCA $= 287,3$ = 288 tiles /<i>teëls</i> ✓CA Number of boxes/ <i>Getal bokse</i>: $288 \div 4$ ✓MCA $= 72$ ✓CA OR/OF Number of tiles/<i>Getal teëls</i> = $4\text{ m} \div 0,35 \approx 11,428$ ✓C MCA $\approx 11,428$ ✓A Number of tiles/<i>Getal teëls</i> = $5\text{ m} \div 0,35 \approx 14,2857$ Total number of tiles for lounge <i>Totale getal teëls vir woonkamer</i> $= 11,4285 \times 14,285 = 163,2641$ ✓SF Number of tiles/<i>Getal teëls</i> = $3\text{ m} \div 0,35 = 8,5714$ Number of tiles/<i>Getal teëls</i> = $4\text{ m} \div 0,35 = 11,4285$ Total number of tiles for dining <i>Totale getal teëls vir eetkamer</i> $= 11,4285 \times 8,5714 = 97,9582$ Total for lounge and dining room <i>Totaal vir woon en eetkamer</i> $= 163,2641 + 97,9582 = 261,22$ tiles ✓CA Including extra for cuttings and breakages/ <i>Insluitend ekstra vir sny en breek</i> $= 261,28 \times 110\% = 287,408$ ✓MCA ✓CA $\approx 287,408$ ✓MCA Total number of boxes/<i>Getal bokse</i> = $287,408 \div 4$ $= 71,852$ ≈ 72 ✓CA	OR/OF 1A simplification 1A factor 1C conversion 1MCA dividing areas 1CA simplification 1MCA calculation 10% 1CA simplification 1MCA dividing by 4 1CA rounded up simplification OR/OF 1C conversion 1MCA dividing dimensions 1A simplification 1 SF substitution 1CA simplification 1MCA calculation 10% 1CA simplification 1MCA dividing by 4 1CA rounding up	(9)

Q/V	Solution/Oplissing	Explanation/Verduideliking	T/L
3.2.3 TR	Bags of tile cement/Sakke teël sement $= \frac{32}{3} = 10,7 \approx 11 \quad \checkmark A$ Cost of the cement/Sementkoste $= R99,90 \times 11 = R1\,098,90 \quad \checkmark MCA \quad \checkmark CA$ Cost of the grout/Koste van bryvulsel $= R89,90 \times 4 = R359,60 \quad \checkmark CA$ Cost of the tiles/Teëlkoste $= R143,84 \times 72 = R10\,356,48 \quad \checkmark CA$ Total cost/Totalekoste $= R10\,356,48 + R1\,098,90 + R359,60 + R2\,500 \quad \checkmark MCA$ $= R14\,314,98 \quad \checkmark CA$ Her budget is enough./Haar begroting is genoeg. $\checkmark O$ OR/OF (using 73 boxes of tiles) Bags of tile cement/Sakke teël sement $= \frac{32}{3} = 10,7 \approx 11 \quad \checkmark A$ Cost (in rand)/Koste in rand $= 143,84 \times 73 + 99,90 \times 11 + 89,90 \times 4 + 2\,500 \quad \checkmark MCA$ $= R10\,500,32 + R1\,098,90 + R359,60 + R2\,500 \quad \checkmark CA \quad \checkmark CA \quad \checkmark CA \quad \checkmark MCA$ $= R14\,458,82 \quad \checkmark CA$ Her budget is enough./Haar begroting is genoeg. $\checkmark O$ OR/OF	CA from Q3.2.2 1A number of bags of cement 1MCA multiplying cost with number 1CA cement cost 1CA grout cost 1CA tile cost 1MCA adding 4 values 1CA simplification 1O verification OR/OF 1A number of bags of cement 1MCA multiplying cost with number 1CA cement cost 1CA grout cost 1CA tile cost 1MCA adding 4 values 1CA simplification 1O verification 3 marks cement cost 1 mark tile cost 1 mark grout cost 2 marks adding costs 1 mark verification	M/F L4

Q/V	Solution/Oplissing	Explanation/Verduideliking	T/L
	OR/OF Bags of tile cement/Sakke sement $\frac{32}{3} = 10,7 \approx 11 \quad \checkmark A$ Budget verification/Begroting verifikasie: $\checkmark MCA \quad \checkmark CA \quad \checkmark CA$ $R15\,000 - [(R143,84 \times 72) + (4 \times R89,90) + (11 \times R99,90) + R2\,500]$ $\checkmark MCA$ $= R15\,000 - (R10\,356,46 + R359,60 + R1\,098,90 + R2\,500)$ $= R15\,000 - R14\,314,98$ $= R685,02 \quad \checkmark CA$ $\checkmark O$ The budget is enough with R685,02 to spare Die begroting is genoeg met R685,02 oorblywend.	OR/OF 1A number of bags of cement 1MCA multiplying cost with number 1CA tile cost 1CA cement cost 1CA grout cost 1MCA adding 4 values 1CA simplification 1O verification (8)	
		[29]	

QUESTION/VRAAG 4 [30 MARKS/PUNTE]			
Q/V	Solution/Oplissing	Explanation/Verduideliking	T/L
*4.1.1	Right/Regs ✓✓ RT	2RT correct direction (2)	MP L1
4.1.2 (a)	✓ RT K 11 ✓ RT	1RT correct row 1RT correct seat (2)	MP L2
*4.1.2 (b)	Total seats/Totale sitplekke = $10 + 16 \times 5 + 19 + 21 = 130$ ✓ A Ratio/Verhouding = $4 : 130$ ✓ MCA = $2 : 65$ ✓ CA OR/OF Total seats/Totale sitplekke = $64 + 66$ (vacant) = 130 ✓ A Ratio/Verhouding = $4 : 130$ ✓ MCA = $2 : 65$ ✓ CA	1A total seats 1MCA ratio in correct order 1CA simplification OR/OF 1A total seats 1MCA ratio in correct order 1CA simplification (3)	MP L2
4.1.3	Total vacant seats/Totale oop sitplekke = 66 ✓ A Percentage income lost/Persentasie inkomste verloor = $\frac{66}{130} \times 100\%$ ✓ MCA = 50,76923077 ≈ 50,77 % ✓ CA OR/OF Percentage income from occupied seats Persentasie inkomste van hierdie sitplekke = $\frac{64}{130} \times 100\% \approx 49,23\%$ ✓ CA ✓ MCA Income lost/Verlore inkomste = $100\% - 49,23\%$ = 50,77% ✓ CA	CA Q4.1.2 total seats 1A total vacant seats 1MCA percentage calculation 1CA simplification OR/OF 1CA % occupied seats 1MCA subtracting from 100% 1CA simplification NPR (3)	MP L3

Q/V	Solution/Oplossing	Explanation/Verduideliking	T/L
4.2.1	✓✓ O To allow air inside the tank to escape as the diesel is pumped in OR Release air bubbles formed OR To let air in OR To protect the tank from exploding or imploding <i>Om lug wat in die tenk is, uittelaat terwyl diesel ingepomp word</i> OF <i>Lugborrels vry te laat</i> OF <i>Om lug in te laat</i> OF <i>Om te keer dat die tenk ontplof of inplof.</i>	2O reason (2)	MP L4
4.2.2(a)	Inner diameter/<i>Binne-middellyn</i> ✓ A $= 3,22 \text{ m} - 2 \times \frac{5}{1\,000} \text{ m}$ $= 3,21 \text{ m}$ OR/OF ✓ A ✓ C $5 \text{ mm} + 5 \text{ mm} = \frac{10}{1\,000} \text{ m} = 0,01 \text{ m}$ $\text{Inner Diameter} = 3,22 - 0,01 \text{ m}$ $= 3,21 \text{ m}$	1A subtracting double the thickness 1C converting to m OR/OF 1A subtracting double the thickness 1C converting to m (2)	M L2
4.2.2 (b)	Inner height/<i>Binne hoogte</i> ✓ MA $= 7,25 \text{ m} - 2 \times \frac{5}{1\,000} \text{ m}$ $= 7,24 \text{ m}$ ✓ CA ✓ MCA $\text{Volume} = 3,142 \times \left(\frac{3,21}{2}\right)^2 \times 7,24$ ✓ SF $= 3,142 \times (1,605)^2 \times 7,24$ $= 58,599622782 \text{ m}^3$ ✓ CA Filling volume/<i>Opvul volume</i> $= 58,599622782 \text{ m}^3 \times 95\%$ ✓ MCA $= 55,6696416429 \text{ m}^3$ ✓ CA Number of litres/<i>Hoeveelheid liter</i> $= 1\,000 \times 55,6696416429 \text{ m}^3$ $= 55\,669,64 \text{ l}$ ✓ C	1MA subtracting double the thickness 1CA simplification 1MCA finding radius 1SF correct values 1CA simplification 1MCA percentage finding 1CA capacity 1C to litres (8)	M L3

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
4.2.3	$SA/BO = 2 \times 3,142 \times (1,61) \times (1,61 + 7,25)$ $\approx 89,64 \text{ m}^2$ Total area to be painted/<i>Totale oppervlakte om te verf</i> $= 89,64 \text{ m}^2 - 1 \text{ m}^2$ $= 88,64 \text{ m}^2$ Litres needed/<i>Liter nodig</i> = $88,64 \div 3$ $= 29,55$ Valid OR/OF $SA/BO = 2 \times 3,142 \times (1,61) \times (1,61 + 7,25)$ $\approx 89,64 \text{ m}^2$ Surface Area = $89,64 \text{ m}^2 - 1 \text{ m}^2$ $= 88,64 \text{ m}^2$ Area that can be covered by 30 ℓ /<i>Opp wat met 30 ℓ geverf word</i> $30 \text{ litres} \times 3 = 90 \text{ m}^2$ Less is needed/ <i>Minder word benodig</i>	CA from Q4.2.2 1MCA correct radius 1SF substitution 1S simplification 1MCA subtracting 1 m^2 1CA simplification 1MCA dividing by 3 1CA simplification 1O verification OR/OF 1A correct radius 1SF substitution 1S simplification 1MCA subtracting 1 m^2 1CA simplification 1MA multiplying by 3 1CA simplification 1O verification	M L4 (8) [30]

QUESTION/VRAAG 5 [28 MARKS/PUNTE]			
Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
5.1.1	✓✓ RT Front view OR Back view OR Rear view <i>Vooraansig OF Agteraansig</i>	2RT view (2)	MP L1
5.1.2	Width of the bakkie/Bakkie se breedte = 1,86 m ✓ C $2D = 3,6 \text{ m} - 1,86 \text{ m}$ $= 1,74 \text{ m}$ ✓ MA $D = \frac{1,74}{2} \text{ m}$ ✓ MCA $= 0,87 \text{ m}$ ✓ CA OR/OF Width of the garage/Motorhuis se breedte = 3 600 mm ✓ C $2D = 3\,600 \text{ mm} - 1\,860 \text{ mm}$ $= 1\,740 \text{ mm}$ ✓ MA $D = \frac{1740}{2} \text{ mm}$ ✓ MCA $= 870 \text{ mm}$ ✓ CA	1C conversion 1MA difference 1MCA dividing by 2 1CA simplification OR/OF 1C conversion 1MA difference 1MCA dividing by 2 1CA simplification (4)	M L2
5.1.3	Number of choices/Getal keuses = 4×2 ✓ MA $= 8$ ✓ CA	1MA multiplying 1CA number of choices. (2)	P L2
5.2.1	A map is drawn to scale while a strip chart is not. ✓✓ A <i>'n Kaart word volgens skaal geteken terwyl 'n strook kaart nie.</i> OR/OF A map shows the routes in a winding manner while a strip chart shows them as straight lines. <i>'n Kaart toon die kronkelende roetes terwyl die strook kaart dit in reguitlyne wys.</i>	2A statement (2)	MP L1

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
*5.2.2	Distance/ <i>Afstand</i> (Springbok to/na Gobabis) $\checkmark$ RT $= 892 \text{ km} + 203 \text{ km}$ $\checkmark$ RT $= 1\,095 \text{ km}$ $\checkmark$ CA	1RT correct 892 1RT adding 1CA distance in km (3)	MP L2
5.2.3	Noordoewer $\checkmark\checkmark$ RT	2RT correct town (2)	MP L2
5.2.4 (a)	Distance Mariental to Keetmanshoop <i>Afstand van Mariental na Keetmanshoop</i> $\checkmark$ RT $= 644 - 427 = 217 \text{ km}$ $\checkmark$ A Total distance travelled/ <i>Totale afstand afgelê</i> $= 140 \text{ km} + 289 \text{ km} + 217 \text{ km} = 646 \text{ km}$. $\checkmark$ CA OR/OF Distance/ <i>Afstand</i> $= 140 \text{ km} + 289 \text{ km} + (465 \text{ km} - 248 \text{ km})$ $\checkmark$ RT $= 140 \text{ km} + 289 \text{ km} + 217 \text{ km}$ $\checkmark$ A $= 646 \text{ km}$ $\checkmark$ CA	1RT distances 1A simplification 1CA distance OR/OF 1RT distances 1A simplification 1CA distance (3)	MP L2

Q/V	Solution/Oplissing	Explanation/Verduideliking	T&L
5.2.4 (b)	Time/Tyd 1 = $140 \text{ km} \div 80 \text{ km/h} = 1,75 \text{ hrs}$ ✓ SF ✓ S Time /Tyd 2 = $289 \text{ km} \div 80 \text{ km/h} = 3,6125 \text{ hrs}$ ✓ S Time/Tyd 3 = $217 \text{ km} \div 120 \text{ km/h} = 1,808333333 \text{ hrs}$ ✓ S Stoppage time = $3 \times 25 \text{ min} = 75 \text{ min} = 1,25 \text{ hrs}$ ✓ S Travelling time including breaks = $1,75 + 3,6125 + 1,808333333 + 1,25$ ✓ MCA = $8,420833333 \text{ hrs}$ ✓ CA = $8 \text{ h } 25$ ✓ C Travelling time = $12:25 - 04:00$ ✓ MA = $8 \text{ h } 25$ ✓ A Letitia's statement is CORRECT/KORREK ✓ O OR/OF Total time taken/Totale tydsduur = $12:25 - 4:00$ ✓ MA = $8 \text{ h } 25 \text{ min}$ ✓ A Driving time on gravel road/Bestuurstyd op grondpad = $\frac{429 \text{ km}}{80 \text{ km/h}}$ ✓ S ✓ SF = $5,3625 \text{ h}$ ✓ S Driving time on tarred road/Bestuurstyd op teerpad = $\frac{217 \text{ km}}{120 \text{ km/h}}$ = $1,808333 \text{ h}$ ✓ S Total time/Totale tyd = $5,3265 \text{ h} + 1,808 \text{ hr}$ = $7,170833 \dots \text{ hours/uur}$ ✓ CA = $7 \text{ hours} + 0,17083333 \times 60$ = $7 \text{ h } 10 \text{ min}$ ✓ C ∴ Total break time/Totale rustyd = $8 \text{ h } 25 \text{ min} - 7 \text{ h } 10 \text{ min} = 1 \text{ h } 15 \text{ min}$ ✓ CA Duration of breaks/Rustye se duur = $3 \times 25 \text{ min}$ = 75 min = $1 \text{ h } 15 \text{ min}$ ✓ A Each break/Elke rustyd = $\frac{1 \text{ h } 15 \text{ min}}{3}$ = 25 min Letitia is CORRECT/KORREK ✓ O OR/OF	1SF substitution 1S simplification 1S simplification 1S simplification 1S simplification 1MCA adding time 1CA simplification 1C converting time 1MA subtracting 1A total travelling time 1O opinion OR/OF 1MA subtracting 1A total travelling time 1S total distance 1SF substitution 1S simplification 1S simplification 1CA simplification time 1C converting time 1CA simplification 1A break time 1O opinion OR/OF	M L4

Q/V	Solution/Oplossing	Explanation/Verduideliking	T&L
	$\checkmark$ SF $\checkmark$ S Time/Tyd 1 = $140 \text{ km} \div 80 \text{ km/h} = 1 \text{ h } 45 \text{ min}$ Time /Tyd 2 = $289 \text{ km} \div 80 \text{ km/h} = 3 \text{ h } 36 \text{ min}$ $\checkmark$ S Time/Tyd 3 = $217 \text{ km} \div 120 \text{ km/h} = 1 \text{ h } 48 \text{ min}$ $\checkmark$ S Travelling time/Reis tyd = $1 \text{ h } 45 \text{ min} + 3 \text{ h } 36 \text{ min} + 1 \text{ h } 48 \text{ min}$ $\checkmark$ MCA = $7 \text{ h } 9 \text{ min}$ $\checkmark$ CA Travelling time /Reis tyd = $12:25 - 04:00$ $\checkmark$ MA = $8 \text{ h } 25 \text{ min}$ $\checkmark$ A $\therefore$ Total break time/Totale rustyd = $8 \text{ h } 25 \text{ min} - 7 \text{ h } 9 \text{ min} = 1 \text{ h } 16 \text{ min}$ $\checkmark$ CA Each break/Elke rustyd = $\frac{1 \text{ h } 16 \text{ min}}{3}$ $\approx 25 \text{ mins}$ $\checkmark$ S Letitia's statement is CORRECT/KORREK $\checkmark$ O	1SF substitution 1S simplification 1S simplification 1S simplification 1MCA adding time 1CA simplification 1MA subtracting 1A traveling time 1CA simplification 1S break time 1O opinion (11)	
		[29]	

NOTES: MATHEMATICAL LITERACY PAPER 2**Level 4 Questions: Calculations must be evident to award the conclusion/opinion mark.****When rounding it must be correctly rounded to a minimum of 2 decimal places unless stated otherwise.****On higher order (i.e level three to four multi-step calculations) questions no penalty for correct early rounding.****QUESTION 1**

1.1.4 Accept: B

1.1.5 Accept: E or B

1.2.2 Accept **cubic centimeters** (i.e. **cm³**) / Kubieke centimeter1.2.4 **CA only apply if 1 value is correct.** That is, either 2 100 or 70 must have been used in a fraction for a **max. 2 marks**, on condition it is correctly simplified.1.3.5 Accept, for **full marks** description:

- Ten past two in the afternoon. / *Tien oor twee in die namiddag.*
- Ten past two post meridian. / *Tien oor twee meridiaan*
- Ten past two pm / *Tien oor twee nm*

QUESTION 2

2.2.2 Accept East of South

2.2.3 Accept one of the following street names for full marks:

- King.
- Pioneer.

QUESTION 33.2.1 Candidates need not show $(20 + 12)m^2$ 3.2.2 **Full marks can be awarded for this solution:****Lounge:** Length = $4m \div 0,35$

= 11,428

 ≈ 12 Width = $5m \div 0,35$

= 14,285

= 15

□ Total tiles = 12×15

= 180 tiles

Dining: Length = $4m \div 0,35$

= 11,428

 ≈ 12 Width = $3 \div 0,35$

= 8,571

 ≈ 9 □ Total tiles = 11×9

= 108 tiles

Hence, total tiles needed = $180 + 108$

= 288

Number to add = $288 \times 1,1$

= 316,8

 ≈ 317 □ Number of boxes = $317 \div 4$

= 79,25

 ≈ 80 boxes

QUESTION 4	
4.1.1	Accept: • East / Oos or E / O
4.1.2 (b)	Accept, for full marks ratio given as: • 4:130 or $\frac{4}{130}$ However, if given 4:incorrect 2nd part. Did not show how incorrect 2nd part was obtained can get max. 2 marks provided it is simplified correctly. Accept answer simplified into unit ratio.
QUESTION 5	
5.2.2	CA considered only if adding distance from strip chart other than 203km, then (max 2 marks).